

Artificial Sense of Slip - A Review

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Abstract—Slip sensing is important in robotic dexterous manipulation and in advanced upper limb prosthetics because it provides useful information for conveniently adapting manipulation forces. In this paper the physical phenomena involved in slip occurrence are briefly examined, as well as the physiological bases of the human “sense of slip.” Transduction principles and technological approaches, exploited over the years for reproducing the “artificial sense of slip,” are analyzed, with the final aim of identifying the most relevant open issues, as well as research trends.

Index Terms—Tactile sensors, slip sensing, robotic manipulation, prosthetics.

I. INTRODUCTION

SINCE the 16th century, when Ambroise Paré presented the first anthropomorphic prosthetic hand with moving fingers actuated by spur gears and levers [1], the progress in the field of upper limbs prostheses, and—more recently—also in the field of robotic dexterous manipulation, has been steady [2], fostered by the societal need of providing amputees with prostheses mimicking esthetically and functionally the human hand. Human manipulation capabilities, exploited in about 60% of daily life activities [3], are still unmatched. To get close to the performance of their biological counterpart, hand prostheses should allow the manipulation of objects even without additional (e.g., visual) cues, and the effective regulation of both force and precision grasps [4]. To this aim, a sufficient sensory feedback is mandatory [5], [6].

Slip detection plays a central role in grip force adjustment, as well as in manipulation and tactile exploration [5], [7], by providing information about the object properties, e.g. roughness [8], [9], hardness [10] and shape [11]. Slip sensors employed in the automotive field (e.g., in anti-lock braking systems-ABS, traction control) and in industrial robotics (assembly grippers [12]) are generally not suitable for prosthetics [13]. Therefore, over the years additional strategies have been investigated, as demonstrated by an increasing number of related publications (Fig. 1). But, although the idea of embedding slip sensors in prosthetic hands dates back to the late 1980s [14], artificial tactile sensors are still far from paralleling their biological counterpart [15], [16].

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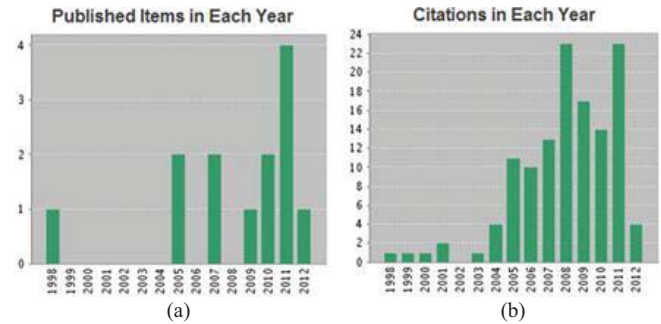


Fig. 1. (a) Number of publications and (b) citations (by ISI Web of Knowledge); research keywords: “Slip sensor prosthetic” or “slide sensor prosthetic” or “slippage sensor prosthetic.”

Slip sensors can be considered as a sub-set of tactile sensors. While valuable review papers on tactile sensors are periodically proposed in literature [13], [16]–[18], to the best of our knowledge this is the first attempt to provide a comprehensive review of slip sensing technologies.

The paper is organized as follows: the physical phenomenon of slip and the mechanisms behind human “slip sense” are briefly described in Sections II and III, respectively. Section IV proposes a synoptic taxonomy of state-of-the-art, which is then analyzed and discussed in Section V. Finally, in Section VI conclusions highlight research trends in the field.

II. BASIC DEFINITIONS

The contact between two bodies, say a finger pad (P) and an object (O), occurs over small surfaces, sometimes approximated to points, placed over an ideal contact surface, S . To avoid interpenetration or detachment, any couple of coincident points $(p, q) \in S$, with $p \in P$, $q \in O$, must move with the same speed in the direction perpendicular to S . Slip occurs if the two points have different tangential velocities.

Incipient (or micro) slip is a relative displacement occurring in a narrow region of S , while total (or macro) slip involves any point of S . These definitions imply that incipient and total slips are temporally contiguous phenomena, with incipient slip preceding total slip.

Let’s consider the interaction forces at the contact surface, as schematized in Fig. 2. Here, two fingers hold an object with mass M . Limiting the analysis to the vertical direction, a stable grip requires the compression forces to be regulated so that $F_{t1} + F_{t2} = Ma$. Of course, according to Coulomb’s friction model, $F_{tk} \leq \mu_s F_{nk}$ ($k = 1, 2$), where μ_s is the (static) friction coefficient. The simplest strategy to prevent object slipping consists in regulating the normal component of the grip force (F_n) to its least effective value. Indeed, slip events

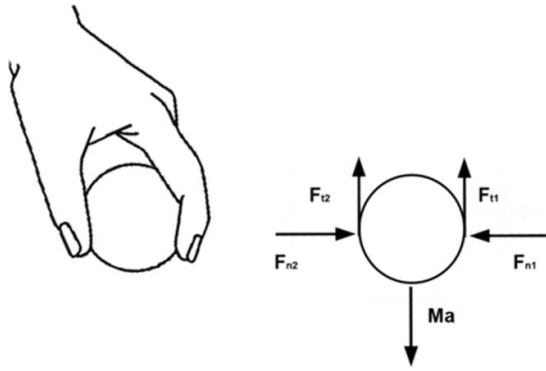


Fig. 2. Grasping forces during weight lifting.

may occur if at least one of the following conditions is verified: 1) the normal grip force is insufficient, as it may happen during its gradual application starting from a no-load condition or if object properties (e.g., mass) are wrongly estimated; 2) when the tangential force, which should be counterbalanced, is higher than expected or quickly increasing (e.g., due to an unexpected perturbation, such as a collision); or 3) the friction coefficient becomes too low, as in the case of wet and slippery surfaces.

A direct measurement of slip is therefore useful for regulating handling and grasping forces.

III. NEUROPHYSIOLOGY OF SLIP PERCEPTION

The inner physiological mechanisms of what can be referred to as “slip sense” are still object of investigation and different research paths have been explored during the years. For example, some studies put emphasis on the biomechanical behavior of the hand during manipulation [19]–[21]. A human finger pad is usually modeled as a compliant convex surface. Proprioceptors detect limb posture and movement, while mechanoreceptors, branching within the contact area at different skin depths, mediate the response to mechanical stimuli [22] (Table I) and provide information about friction [23] and interaction forces [24]. Friction is influenced by several tribological factors, including sweat secretion and contact occlusion [25].

Each mechanoreceptor (Table I) encodes the spatiotemporal tactile information as spikes of action potentials, i.e. voltage pulses generated when the stimulus overcomes a threshold. Examples of mechanoreceptors recordings are reported in [26] and [27]. Multiple parallel pathways are used to process and transform somatosensory information before it reaches the Central Nervous System (CNS). Studies dated back to 1920 [28] report that the relative timing of the first impulses elicited in individual units in response to external stimuli are fundamental to the rapid encoding and transfer of tactile information (e.g., direction of applied force and shape of the object surface) [26]. Another crucial issue is related to the decoding of such information. According to the coincidence detection hypothesis [27] central neurons preferentially respond when receiving synchronous inputs from different sources. Once the information is decoded, motor outputs, corresponding to corrective responses, are elaborated according to strategies

TABLE I
SCHEMATIC OF MECHANORECEPTORS TYPES AND RELATED FEATURES

Receptor Type (mm)	Field Diameter (mm)	Spatial Density on the Fingertip (cm^{-2})	Frequency Range (Hz)	Detected Parameter
Meissner (FAI)	3–4	1500	10–60	Skin stretch
Pacinian (FAII)	>20	75	50–1000	Vibration
Merkel (SAI)	3–4	750	0–30	Compressive stress
Ruffini (SAII)	>10	75	0–15	Skin stretch direction

resembling finite-state control algorithms, i.e. based on basic logic rules (e.g., AND, IF-THEN) [27].

The capability to perceive incipient slip is important for the stability of manipulation tasks [29]. The loads over the contact area on the finger pad are not uniform, being higher at the center and lower at the boundary. Therefore, incipient slip starts near the edge of the contact area. It can be easily detected if the surface is characterized by a regular and homogeneous texture (e.g., dots), generating detectable skin vibrations [30]. There are evidences that slip is mainly detected by FAII receptors, the mechanoreceptors most sensitive to vibrations (Table I). The responses of FAII afferent receptors to initial contact are the most important in encoding the intensity of friction forces. Information about slip direction comes from FAI and from “slow acting” receptors (SAI and SAI). Epidermal ridges enhance the tactile spatial acuity by transmitting magnified signals from skin surface to the mechanoreceptors [20]. The spatial resolution for slip detection is about 1 mm [20].

Healthy people adapt forces on the basis of the frictional conditions. The response to external stimuli is generally affected by an accumulation of delays related to: 1) stimulus transduction by receptors; 2) neuronal conduction from the periphery to the central processing unit; 3) elaboration of the response; and 4) muscle activation. The sensorimotor system reacts faster to unexpected sensory events during fine manipulation. This improvement in terms of reaction times is evidently not attributable to an improved performance of sensorimotor or effector mechanisms. Rather, stimuli from mechanoreceptors are managed by: 1) the CNS to control manual tasks mainly by means of predictive strategies based on (forward) internal models [31], [32] (Fig. 3); and 2) unconscious reflex actions via the spinal cord acting more rapidly than voluntary corrective actions [33]. Visual cues about the shape of the object and/or sensorimotor memory of previous, expectedly similar, manipulation experiences can provide the information required to elaborate predictions. A mismatch between predicted and actual sensory information triggers corrective actions in less than 100 ms [17], [34]. Repeated interactions (e.g., grasping attempts) become necessary in case of misleading cues, or when manipulated objects have unusual properties affecting the interaction, such as extremely slippery surface, as in the case of a wet bar soap [35].

Some brain imaging studies identified the cerebral areas involved in grasp and manipulation control. Evidences suggest

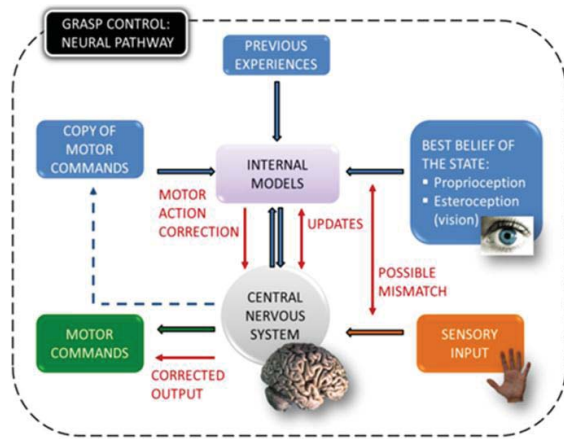


Fig. 3. Neural pathway of grasp stability control.

that the right posterior parietal cortex, as well as the bilateral cerebellum, play important roles, possibly storing the forward internal models [36]. However, force adaptation (i.e. basic grip/load force coupling) seems to be robust to a variety of cerebral lesions, including cerebellar ones [36], which suggest that the central neural architecture supporting these skills is still not exactly identified. What is demonstrated is that humans control grasp forces almost optimally, with forces just 10–40% above the minimum value required to prevent slip [37].

IV. ARTIFICIAL SLIP SENSORS: A TAXONOMY

The basic slip sensor is a binary device able to detect slip absence. Some sensors detect slip at its incipience, other only total slip. More elaborated sensors can provide information about slip direction and speed. Even if most of tactile sensors are bioinspired [17], [18], humans “slip sense” is hard to emulate because there is not a single specific receptor devoted to slip detection to be mimicked.

The lack of a clear biological model to take inspiration from generated a varied landscape of approaches, that differ from each other in terms of: 1) parameters to be monitored (e.g., object displacement, microvibrations, normal and shear forces, thermal flows); 2) transduction mechanisms (e.g., piezoresistive, piezoelectric, capacitive, optical, magnetic, thermal); and 3) technological solutions (e.g., soft/flexible materials, MEMS).

A possible taxonomy of slip sensors is shown Fig. 4, where four main classes of sensors are proposed according to the physical quantity monitored. Each class encompasses sub-categories characterized by different transduction means. Specific devices in the sub-categories differ for their technological implementation.

A. Displacement Sensors

Displacement-based slip sensors detect the relative motion between finger tips and the object (Fig. 5).

Such displacement may cause the motion of mechanical parts, e.g., discs or balls, connected to magnetic or optical transducers [38]. Since mechanical parts are prone to the

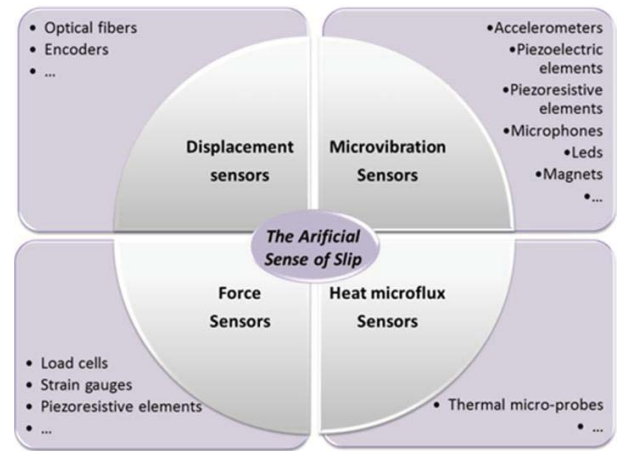


Fig. 4. Slip sensors: a taxonomy based on the detected physical quantity associated to slip.

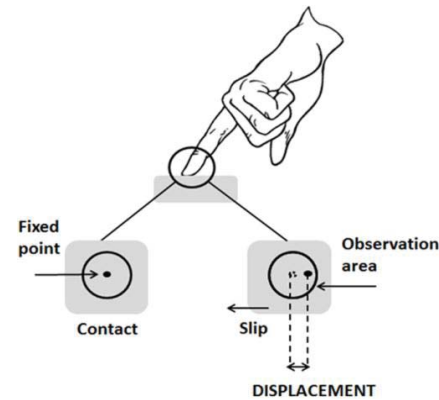


Fig. 5. Slip detection by displacement monitoring. Slip causes the motion of a point within the contact area.

accumulation of impurities, more recent sensors have no external moving mechanism, and detect relative displacement by comparing a sequence of images captured by a CMOS sensor at each flashing of a light-emitting diode (LED). Both mechanical and optical embodiments may provide information about slip speed and direction. This class of sensors has been mainly mounted on the end-effectors of industrial manipulators. Despite their low cost and insensitivity to object roughness, their application to hand prostheses is hindered by their difficult integrability into cosmetic gloves. Moreover, moving mechanical parts, mediating the contact between the robotic finger pad and the object, may negatively affect grasp stability. A few attempts have been reported integrating optical and pressure sensors. The data flow generated by these multimodal sensors was managed by bioinspired algorithms [39], [40].

The integration of a vision system and a displacement tactile sensor is computationally demanding, requiring both image processing and soft computing (e.g., Hebbian artificial neural networks [39]) algorithms. The high computational cost, nonetheless, is compensated by the good reliability of the whole slip sensor, deriving from its intrinsic redundancy [39].

B. Microvibration Transducers

The onset of slip produces small vibrations at the contact surface between the fingertip and the object (Fig. 6). In humans the “slip sense” is mainly based on detecting vibrations ranging from 50 to 1000 Hz (FAII receptors). Microvibrations detection has been the most pursued approach so far in bioengineering. Evidences have also been provided about the capability of this strategy of detecting incipient slip [17], [18].

Vibrations have multiple harmonic components, with different frequencies and phases. Theoretically, vibrations can be monitored by measuring the displacement of a seismic mass and calculating its first and second time derivatives. Displacement measures emphasize the amplitude of low frequency harmonics (< 1 Hz), while acceleration emphasizes high frequencies (> 100 Hz).

Vibration can be detected directly (by measuring the intensity and frequency of the vibration) or indirectly (by characterizing the excitation source, i.e. forces/torques applied to the vibrating system).

Direct vibration transducers are accelerometers or acoustic microdevices. Indirect transducers are usually based on load sensors. In both cases, due to the nature of the signal, a Fourier analysis is usually performed. Different algorithms [41] are developed for scanning the FFT signal. The analyzed parameters can then be compared with a proper threshold value, using an offline elaboration [41], [42], or they can be evaluated online using an artificial neural network [42].

Accelerometers: Accelerometers are a well-established sensor technology, characterized by a small footprint and low power consumption. Over the years they found several applications in hand prosthetics. Pioneering works in the field of slip sensing resorted to one single accelerometer embedded in a robotic fingertip and covered by a foam [43], [44]. The detection technique has been later improved by adding a second accelerometer for enhancing the signal-to-noise ratio [44]. In line with the same approach, more recently three-axis accelerometers have also been used [45].

Accelerometers have also been integrated in sensorized skins as part of a network of multimodal sensor modules. In the HEX-O-SKIN [46], impact, slip, and roughness are inferred from a cost-effective, small-size, and low-power digital 3-D accelerometer (BOSCH BMA150, also embedding a temperature sensor). This solution, currently applied to humanoid robots [46], looks plausibly applicable to prosthetics.

Acoustic Detectors: MEMS microphones can detect the ultrasound (from 50 kHz to 1 MHz) Acoustic Emission (AE) at the contact area associated to slip [47]. AE has been exploited since 1987 in the field of robotic manipulation to detect slip [48]. Unfortunately, acoustic transducers are sensitive to external disturbances [49]. In order to improve the signal-to-noise ratio, the microphone can be isolated suspending it in a foam-filled chamber contacting to the finger surface through an elastomeric membrane [49]. In this way the microphone is mechanically decoupled from the external environment as well as from the possible vibrations generated by motors. This detection strategy is highly dependent from object surface

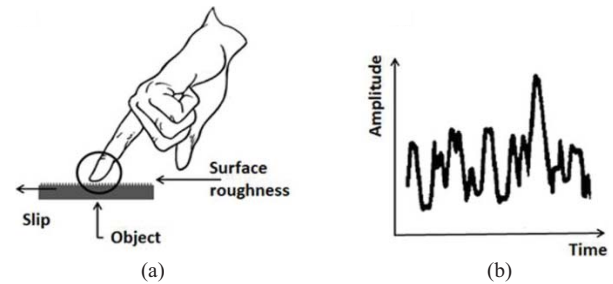


Fig. 6. Slip detection by microvibrations monitoring. (a) Onset of vibrations during slipping due to the roughness of the surfaces in contact. (b) Typical vibration signal.

roughness, though. Shinoda *et al.* in 2000 proposed an acoustic resonant tensor cell (ARTC) including a cavity inside an elastic body and an ultrasound detector. The proposed sensor is capable of detecting slip direction by analyzing the resonant frequency of ultrasonic waves [50].

Recently, a Bayesian exploration algorithm has been developed and implemented using an acoustic slip sensor (BioTac, by SynTouch [51]). The algorithm allows to adaptively select the optimal explorative movements for characterizing and identifying the touched objects [51]. Tests have shown that the exploratory movements allows the slip sensor to be used to discriminate similar textures with a success rate exceeding that of humans.

Piezoelectric Materials: Being capable of directly converting mechanical energy into electric energy, piezoelectric materials have been widely used for detecting slip-induced vibrations [52]. Polyvinylidene Fluoride (PVDF) is a flexible polymer, available in film, with a good dynamic response. PVDF has been used for developing multi element-stress-rate sensors embedded in robotic fingertips made of silicone rubber and foam [53]. PVDF strips can be placed at different depths beneath the surface of a soft robotic fingertip, allowing the detection of the three components of the subsurface stresses [42], [54], [55].

In order to evaluate the friction coefficient, strain sensors have also been used, monitoring the local deformations in the stick/slip regions of the contact area [53], [56].

PZT (Lead Zirconate Titanate) is a ceramic perovskite material, with a sensitivity about one order of magnitude higher than that of PVDF (130 pC N⁻¹ for typical PZT, compared to 20–30 pC N⁻¹ of PVDF) [42]. Also Fluorocarbon PVF2 has been used for dynamic measurements and friction estimation [44]. In order to amplify the reading from the sensor, both cantilever [52] and bimorph arrangements [57] have been used.

Piezoresistive Materials: Piezoresistive materials are largely used in force sensors, but a few examples exist of slip sensors employing these materials [58]. Recently a MEMS-based technology, named Micro-Force/Moment-Sensing (MFMS), has been presented [59]. The sensory system consists of four crossed beams embedding piezoresistors. Experiments have shown that the sensor response time should allow to reach performance comparable to those of humans (i.e. forces adaptation in less than 100 ms) [17], [34].

Photoelastic Materials and Photoelectric Devices: Some photoelastic materials, exhibiting birefringence at low strain, can be exploited in slip sensing [60]. Early prototypes resorted to a camera and dedicated image processing techniques to infer slip. The computational cost of this sensors increases with the desired spatial resolution, which typically is of about 0.1 mm [61].

Gofuku *et al.* (2000), embedded a slip sensor, comprised of photo-reflectors and a conductive rubber, in an artificial hand [62]. Slip detection was limited to objects with a patterned surface (e.g., color or roughness changes over the contact area), thus precluding its effectiveness in case of smooth or monochromatic objects.

In another work, by Chen *et al.* (2010), slip-induced microvibrations were detected by a photoelectric apparatus, comprising LEDs and photodiodes, coupled to a vibrating rubbery sensing element [63]. A fuzzy logic controller successfully modulated grasping forces according to slip signals.

Others: In addition to those already outlined, other, less common, transduction principles have also been exploited over the years.

A Hall vibration detector has been used in [64]. Here, an elastomer, deformed by external forces, displaces a magnet, whose position is detected by the Hall sensor.

Also pressure sensitive conductive rubbers, coupled to Center of Pressure (CoP) tactile sensors, have been employed for robotic hands [65] in order to detect high frequency vibration components due to object slipping.

C. Force Sensors

Force sensors can be used to infer slip occurrence, provided that all force and torque components at the contact point are measured [66] (Fig. 2). Conversely, not all force/torque components are needed if the detection is limited to: 1) vibrations induced by stick-slip phenomena [44]; 2) changes in pressure distribution [41]; or 3) surface deformations [56].

One of the simplest control methods for avoiding slip events is to estimate in real time the normal grip force needed to balance the tangential force (Fig. 2). Such calculation may be directly performed if the friction coefficient is known and it may take into account the mechanical properties of the fingertip through a contact model, e.g. Hertz model [66].

Three-axis force sensors have been the most used, because they allow to simultaneously detect normal and shear forces [67]. Wettels *et al.* in 2009 presented a grasp control algorithm for anthropomorphic prosthetic hands [68], employing a biomimetic tactile sensor based on a compliant three-axis force sensing mechanism [69]. The sensor consists of a rigid central core with electrodes distributed over its surface; a conductive fluid and a silicone elastomeric skin covering the whole system. By applying an alternating current to each electrode, the impedance of each volumetric flow path (from a fixed external point to a reference electrode) can be characterized. Once external forces are applied, the impedance pattern changes according to force magnitude and direction, contact point location and object shape [68]. A similar approach has been implemented using MEMS strain

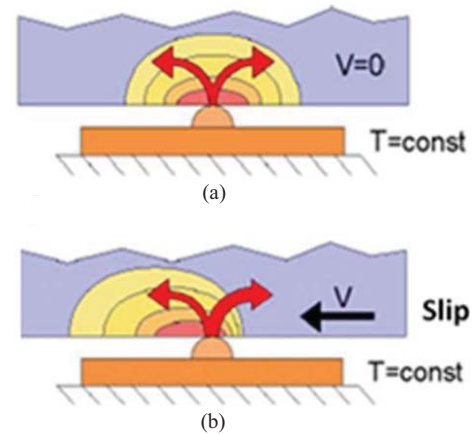


Fig. 7. Slip detection by monitoring thermal changes in thermal microfluxes. (a) When a microheater is in thermal contact with an object, with zero relative speed, only heat conduction occurs. (b) During object motion, also a convective heat transfer occurs.

gauges on a flexible substrate [70]. MEMS technology has also been adopted for developing a 1.4 mm^3 3-D silicon microsensor embedded in a soft, compliant packaging [71], [72]. The microsensor consists of a high aspect ratio silicon structure (625 μm thick), an integrated silicon mesa, transmitting forces to a flexible tethered structure, and four piezoresistors used independently to detect normal and shear contact forces [73]. Slip experiments, focused on the transitions between static and dynamic friction conditions at the finger-object interface, have been performed. Sensor signals have been processed using a cumulative summation algorithm. Results show response times in slip detection between 24.5 and 44 ms [73].

D. Heat Microfluxes Detectors

Thermal sensors have been mainly used in robotics hands for measuring objects thermal properties, such as temperature, heat capacity and thermal diffusivity [74]–[76]. The first example of a thermal slip detector, where the temperature change induced by the motion of the touched object was recorded, was presented in 1986 for the detection of gross slip phenomena in industrial robotics application [77]. Accoto *et al.* presented in 2008 a slip microsensor where the sensing element is a planar gold microheater, kept at a constant temperature, slightly above room temperature (isothermal sensor), by a dedicated bang-bang current control [78]. The current supplied is allowed two values: a lower one (when the microheater is required to cool down) and a higher one (for warming up the microheater). Slip increases the heat flow from the heater (warmer) to the object (cooler), as shown in Fig. 7. A slip signal is fired, typically in less than 60 ms, when the increase of the current needed for keeping the microheater in isothermal condition overcomes a given threshold. The sensor, with no moving parts, is not affected by spurious vibrations and its performance does not depend on object roughness. The first prototype has been later optimized [79] and fabricated on a flexible substrate to ease its integration over curved robotic fingertips [80], [81].

TABLE II
MAIN ADVANTAGES AND DISADVANTAGES OF THE REVIEWED SLIP SENSORS

Transduced Physical Slip-Related Quantity	Transduction Technology	Advantages	Disadvantages
Displacement [38]–[40]	Optical	Low cost components; Possibility of implementing learning procedures based on artificial vision	Bulky readout system; No incipient slip detection; Possible optical path occlusion
Microvibrations	Piezoelectric [44], [52]–[57] Acoustic [47], [49], [50], [80] Piezoresistive [58], [59] Photoelastic [60], [62], [63] Magnetic [64]	Wide dynamic range; Good response time; Simple design	Sensibility and stability limited by technology; Possible damping by elastomeric materials (e.g. skin material); Performance depending from object surface finishing; Affected by external mechanical vibrations
Forces [44], [56], [66]–[73]	Capacitive Conductive elastomers Piezoelectric Piezoresistive Strain gauge	Good sensitivity; Wide dynamic range; additional information provided about object shape and contact conditions (forces/pressure)	Complex; Susceptible to noise
Thermal microfluxes [78]–[81]	Resistance temperature detectors (RTD)	Insensitive to object surface finishing; Robustness with respect to external noise (including mechanical vibrations); Easy manufacturability	Slip occurrence is not distinguished from contact

TABLE III
QUALITATIVE EVALUATION OF SLIP SENSORS; SYMBOLS LEGEND

Transduced Parameter	Power Consumption	Compliance	Sensitivity to External Noise	Manufacturability	Computational Cost	Incipient Slip Detection
Displacement	**	-	*	**	***	-
Microvibration	**	**	**	*	**	***
Force	*	***	*	*	**	**
Thermal microflux	*	***	-	***	*	-

‘-’: none; ‘*’: minor; ‘***’: medium.

V. DISCUSSION

A slip sensor is intended to detect the occurrence or incipience of slip, although it may also provide additional information, such as slip speed and direction. The acknowledgement of the importance of slip sensors in prosthetic devices dates back to the 1970–80s [14]. Since then, the research effort spent on the field is steadily growing (Fig. 1).

In the case of prosthetics applications it is desirable the sensor to be compact (small size and with few parts, for an easier integration), mechanically compliant (i.e. flexible or stretchable to better adapt to curved surfaces, such as fingertips), and robust against electrical and magnetic interferences, spurious vibrations (e.g., caused by moving mechanical parts), impacts, abrasion and wear.

Slip sensors are often integrated into curved and soft fingertips. In fact, it has been demonstrated that the use of compliant fingertips is beneficial for the effectiveness of grasping [82], [83]. As it regards the response time, it should be smaller than that of the human sensors (i.e. $\ll 100$ ms) regardless of

the properties of the grasped object. This requirement arises from the need of allowing sufficient time to the prosthesis control to adapt grasp in a time window compatible with the generation of smooth, human-like motions. Finally, the sensor should not be too power demanding and costly. Even if the need of slip sensing has been addressed since the dawn of robotic manipulation, still there is not a dominant, well-established technology, and the existing ones do not meet all the above mentioned requirements.

For comparison purposes, we summarized in Tables II and III the main advantages/disadvantages of the reviewed technologies, together with their qualitative evaluation.

The observation of the relative displacement of two surfaces in contact (e.g., object and fingertip), is probably the most intuitive approach for detecting the occurrence of slip. However, this strategy does not allow the detection of incipient slip, which, if timely detected, would at least theoretically enable the adaptation of grasping just before any undesired motion of the object occurs. Under this regard, vibrations-based sensors are probably the most promising, although quite complex.

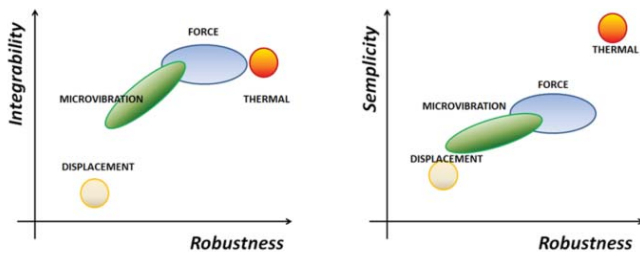


Fig. 8. Comparison of slip detection strategies.

Among vibrations transducers, piezoelectric materials are appealing, because they offer a direct transduction from the mechanical domain into the electrical domain, with minor wiring problems, quick response times and low power consumption. Unfortunately, their performance is affected by object surface properties (e.g., a high surface roughness helps vibrations generation), mechanical noise and damping effects due to the viscoelastic properties of the elastomeric materials usually employed as covers.

Also accelerometers look very promising for the development of incipient slip sensors. The tremendous advancement of MEMS technology in the last two decades made these devices extremely sensitive, compact, reliable and cheap.

As it regards other technologies (force sensors, photoelectric devices and acoustic transducers), a large amount of work should be devoted not just to the development of novel hardware but also to the development of efficient computational techniques able to manage in a very short time the large amount of data produced. The expected important side-benefit, especially in the case of multiaxis force sensor, is related to the capability of the sensor not only to detect (incipient) slip, but also to fully characterize interaction forces at the site of contact. This approach, in a sense, is more bioinspired than others.

As explained in Section III, the sense of slip in humans emerges from the combined analysis of a wealth of different tactile information, collected by several distributed sensors. One might expect that the investigation of artificial multimodal sensory fusion techniques, aiming at detecting incipient slip, may even shed light on some aspect of the neurophysiology of slip sensing in humans.

From a strict engineering viewpoint bioinspiration is not a must. For instance, slip sensors based on thermal phenomena (i.e. detection of convective heat transfer associated to slip) are based on a transduction principle exploited in several technical fields (e.g., hot wire anemometry) but not in nature. Such sensors, requiring a small but finite slip speed, are intrinsically not capable of detecting incipient slip. Nonetheless, they can be easily micro-manufactured using lithographic techniques. Since thermal capacity is directly proportional to mass, response time scales very positively with miniaturization, without any theoretical lower bound. One can expect that research in this field may lead to sensors responding in a few milliseconds (compared to the current 40–50 ms), a time short enough to make the displacement of the handled object before grasp regulation quite small.

In the following we report a qualitative comparison of different transduction strategies for slip detection based on the following performance indices:

Robustness: evaluates the capability of detecting incipient slip rejecting external disturbances;

Integrability: evaluates how simple the integration of the sensor on a robotic fingertip is. This parameter increases if the sensor can be manufactured using flexible or stretchable technologies and diminishes in case, e.g. complex wiring or high power are needed;

Simplicity: increases if manufacturing process and/or computational demands are small.

Fig. 8 shows two bidimensional projections of a 3-D plot where the reviewed technological solutions for slip detection are qualitatively compared in terms of the three defined indices (*Robustness*, *Integrability*, *Simplicity*). Here, force and thermal sensors are considered to be the most promising for application in prosthetic hands.

VI. CONCLUSION

The lack of a rich sensory feedback is one of the current main limits of robotic hands and advanced upper limb prostheses. The human hand is characterized by an unparalleled rich sensory system, which largely contributes to its astonishing manipulation capabilities. In particular, tactile feedback is fundamental in manipulation tasks and slip sensors provide a very basic information, needed to assess the stability of the grip and to trigger corrective measures in case of instability.

In order to amend this lack, especially in the detection of incipient slip, several types of sensors have been proposed over the years, and microtechnologies played a central role in enabling the development of novel components and devices, and in improving their performance, especially in terms of faster response, reduced complexity and encumbrance, and smaller cost. The design of slip sensors is often inspired by human detection/control mechanisms, which are multilevel and integrated: the fusion of sensory information from multiple sources is the key for a robust perception, as it maximizes the amount of information acquired from the (unstructured or unknown) environment. For this reason, it is expected that in the near future we will assist to an increased interest to multimodal array of tactile sensors (including slip detectors), possibly fabricated using compliant (flexible and/or stretchable) packagings [84], [85], which can be better integrated into artificial skins.

In this same direction, the possibility of channeling the sensory information, possibly originated from neuromorphic sensors [86], to a bidirectional neural interface, e.g. exploiting biohybridized microsystems, will enlarge the reach of prosthetics towards the promises of bionics. Although, since 1980s [14], robotic manipulation has been the main application arena of slip sensors, it is important to mention that there are other robotics fields (including haptics and wearable robotics) where slip sensors could enable novel functionalities or improve existing ones.

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